

Civil Engineering

Trigonometry and Complex Numbers

Learner Notes

Trigonometric ratios, sine rule, cosine rule, bearings, and complex numbers.

Learning outcome	Focus
Apply trigonometric ratios	Applied mathematics in workplace and training contexts.
Solve engineering triangle problems	Applied mathematics in workplace and training contexts.
Represent complex numbers in rectangular and polar form	Applied mathematics in workplace and training contexts.

1. Topic Overview

This topic develops mathematical skills required in Civil Engineering. Learners should show clear method, correct units, and technical interpretation.

2. Key Formulas

$$\sin \theta = \frac{\text{opposite}}{\text{hypotenuse}}$$

$$\frac{a}{\sin A} = \frac{b}{\sin B}$$

$$c^2 = a^2 + b^2 - 2ab \cos C$$

$$z = a + bi$$

3. Worked-Example Method

Step 1: identify the given data. Step 2: select the correct formula. Step 3: substitute values carefully. Step 4: compute. Step 5: interpret the answer with correct units.

$$\sin \theta = \frac{\text{opposite}}{\text{hypotenuse}}$$

4. Practice Questions

1. Compute the height of a structure from angle of elevation.

2. Solve a triangular road reserve survey problem.
3. Convert a complex number to polar form.